

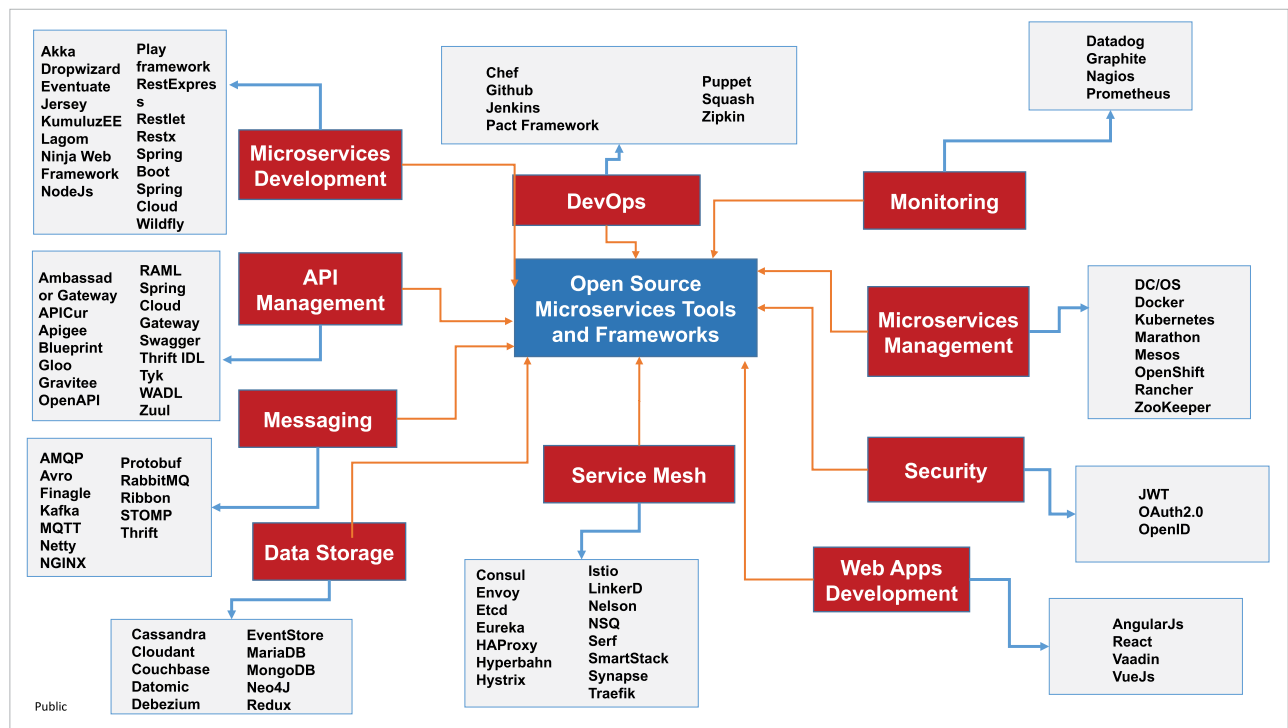


Figure 1: Open source microservices tools and frameworks

work. When one team updates its database structure, other services that use the same structure may also have to be adapted.

Code maturity: All code in a microservice should have a similar degree of maturity and stability. To add or rewrite the code that works well in a microservice, the best way is usually to create a new one for the new or altered code and leave the existing microservice where it is. This helps to iteratively release and test the new code until it is error-free and maximally efficient.

Deployment in containers: Deploying microservices in containers is important; you need one tool to deliver everything. As long as the microservice is in a container, the tool knows how to deliver it. Docker has currently become the standard for containers.

Monitoring: Use logs such as Splunk and monitoring tools like AppDynamics to measure efficiently and simply with mechanisms such as response time notifications, service error notifications and dashboards.

Tools and frameworks for microservices adoption

There exist various tools and frameworks for the development of microservices across the organisation. The major activities of microservices development are:

- API management
- Messaging
- Data storage
- DevOps
- Service mesh
- Monitoring
- Microservices management
- Security
- Web application development

The various features that need to be addressed as part of the microservices development are:

- Load balancing
- Event sourcing
- Asynchronous messaging
- Service discovery
- Service monitoring
- Proxy server
- NoSQL
- CQRS

- CI/CD
- Persistency
- Data provisioning
- Error handling
- API design
- API analytics
- API gateway
- API management
- Authentication
- Container management

Figure 1 summarises the tools and frameworks that need to be adopted in design to deploy the microservices based on open source technologies.

The following are some of the key open source microservices tools and frameworks.

Chef: Chef is an open source automation platform that transforms infrastructure into code. It operates in the cloud, on-premises, or in a hybrid environment. The Chef development kit provides the tools to develop and test infrastructure automation code from the local workstation prior to deploying changes into production.